

MadhuMarga Agri - Project Documentation

Project Title

MadhuMarga Agri

Short Description

A specialized Android application designed for beekeepers to streamline apiary management, including hive registration, health monitoring, and harvest tracking.

GitHub URL

https://github.com/marutigoudra/madhumarga_agri

Problem Statement

Traditional beekeeping often relies on manual or fragmented record-keeping for hive health, inspection cycles, and harvest yields. This makes it difficult to track long-term trends or identify issues across multiple apiaries. MadhuMarga solves this by providing a centralized, digital dashboard to monitor hive lifecycle, floral availability, and productivity in real-time.

Technologies Used

- Android (Kotlin)
- Room Persistence Library (SQLite)
- Jetpack Components (ViewModel, LiveData)
- Material Design
- Gradle (Kotlin DSL)

Key Features

- Hive Management: Digital inventory and registration of hives.
- Inspection Logs: Health tracking, queen status, and pest monitoring.
- Harvest Tracker: Quantitative logging of honey and wax yields.
- Flora Calendar: Tracking local floral cycles for optimized nectar flow planning.
- Offline Support: Fully functional without internet connectivity using Room Database.